

RNE HW3-1 Report

An-Yi Lee
Student ID: S11259029

1 Introduction

This homework implements a PPO agent for the CarRacing environment. The goal is to complete the missing implementations in `model.py`, `env_runner.py`, and `agent.py`, and tune the training hyperparameters in `train.py` to improve the final evaluation score.

2 Implementation

2.1 Model

I implemented the policy network and value network in `model.py`. Both networks use multilayer perceptrons with two hidden layers.

2.2 Environment Runner

I implemented rollout data collection in `env_runner.py`. The runner stores states, actions, log probabilities, values, rewards, and done signals for PPO training.

2.3 PPO Update

I implemented the PPO clipped objective in `agent.py`. During training, advantages are computed from returns and values, and then normalized before performing the PPO update.

In `train.py`, I adjusted the PPO hyperparameters to improve training stability and final evaluation performance.

3 Hyperparameters

The final hyperparameters used in training are:

- `n_env` = 8
- `n_step` = 256
- `sample_mb_size` = 64
- `sample_n_epoch` = 4
- `a_std` = 0.35
- `lambda` = 0.95
- `gamma` = 0.99
- `clip_val` = 0.15
- `lr` = $2e-4$

- `n_iter = 30000`
- `device = cuda` if available, else `cpu`

4 Results

Figure 1 shows the training curve. The reward steadily increased during training and eventually stabilized around 230–240.

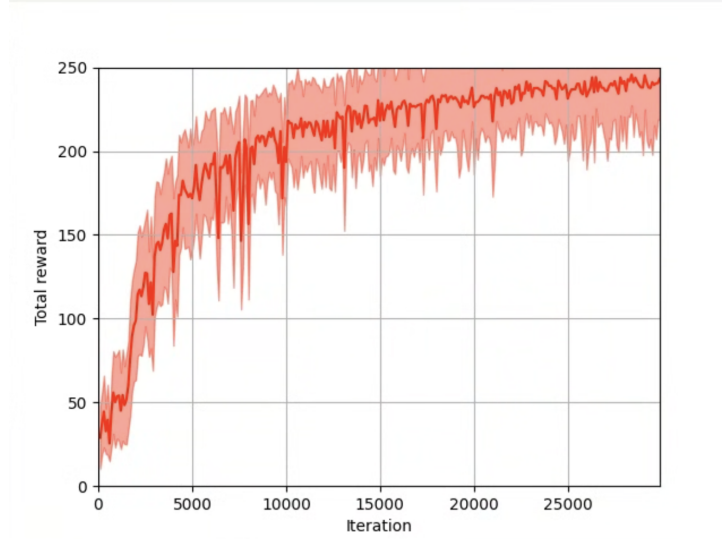


Figure 1: Training reward curve

Figure 2 shows the final evaluation result. The final evaluation score is **259.3584**, which is well above the full-score threshold for HW3-1.

```
Total reward = 234.568302, length = 284
Total reward = 250.843412, length = 299
Total reward = 288.914613, length = 355
Total reward = 286.360077, length = 329
Total reward = 228.764720, length = 295
Total reward = 280.491860, length = 326
Total reward = 236.012865, length = 269
Total reward = 289.341201, length = 323
Total reward = 265.604959, length = 325
Total reward = 290.284297, length = 319
Total reward = 250.388286, length = 285
Total reward = 276.721902, length = 316
Total reward = 258.106724, length = 314
Total reward = 260.219351, length = 324
Total reward = 220.459214, length = 311
Total reward = 283.171684, length = 311
Total reward = 274.947118, length = 328
Total reward = 260.725871, length = 313
Total reward = 249.804202, length = 299
Total reward = 261.700444, length = 291
Total reward = 223.643585, length = 268
Total reward = 271.518950, length = 322
Total reward = 199.293830, length = 366
Total reward = 285.816347, length = 360
Evaluation Score: 259.3584
```

Figure 2: Final evaluation result

5 Conclusion

After fixing the rollout implementation and tuning the PPO hyperparameters, the agent achieved stable training performance and a high evaluation score.